Krishna Aggarwal

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Professional Summary

- Computational neuroscience researcher working in interdisciplinary neuroscience, where AI, mathematics, physics, and computational methods are brought together to understand how cognition arises in the brain.
- Current work draws on dynamical systems, deep learning, and the analysis of neural and behavioural data, with experience spanning simulation, mathematical modelling, and data-driven research.
- Interested in decoding and understanding cognition, decision-making, and the neural mechanisms underlying behaviour, through an interdisciplinary lens that combines biological insights with mathematical and computational principles.
- Motivated to pursue doctoral research at the intersection of theoretical and computational neuroscience, with openness to dynamical-systems modelling, cognitive computational modelling, neuro-inspired AI, and other directions that bring these threads together.

Research Interests

Theoretical & Computational Neuroscience; Cognitive Computational Modelling; Machine Learning for Neural Data; Neural Dynamics & Nonlinear Systems; Dynamical Systems & Mathematical Modelling of Neural Activity; Brain Signal Analysis (EEG, LFP); Neuro-inspired Deep Learning

Publications & Preprints

Aggarwal, K. (2025). Cross-Linguistic Analysis of Memory Load in Sentence Comprehension: Linear Distance and Structural Density. arXiv preprint. https://arxiv.org/abs/2509.20916

This work was accepted for presentation at:

- 7th International Conference on the Mathematics of Neuroscience & AI (ICMNS 2026), Rome, Italy – June 2026
- Dubrovnik Conference on Cognitive Science (DUCOG 2026), Croatia – May 2026

Research Experience

MS Thesis|EEG-based Cursor Navigation with Deep Oscillatory Neural Networks Aug 2025 – April 2026

Computational Neuroscience Lab, IIT Madras | PI: Dr. V. Srinivasa Chakravarthy

- Built Dense Oscillatory Neural Network (DONN) and topomap-based Oscillatory CNN architectures using Hopf-bifurcation oscillator layers, treating temporal dynamics as a learned computational primitive rather than a feature to average away.
- Benchmarked these against EEGNet, CNN-LSTM, CNN with PSD features (XGBoost), and Riemannian (covariance and tangent space) pipelines on 128-channel EEG from 20 subjects, using Leave-One-Block-Out cross-validation for subject-level generalization.
- Designed and ran full data acquisition: custom MATLAB protocol with NetStation SDK integration for millisecond-precision event marker synchronization, plus a preprocessing pipeline with band-pass filtering, ICA-based artefact removal, and epoch-level normalization.
- Studying whether dynamical priors improve robustness and generalization under high trial-to-trial variability, a question relevant to how dynamical structure supports robust neural computation.

Reconstructing STN–LFP Signals Using Neural Networks May 2024 – July 2024

Computational Neuroscience Lab, IIT Madras | PI: Dr. V. Srinivasa Chakravarthy

- Built coupled Rössler oscillator networks with Hermitian complex-weight coupling as a generative model for subthalamic nucleus LFPs from Parkinson's patients, with synchronization, phase coupling, and beta-band dynamics as the core computational primitives.
- Compared feedforward, LSTM, and Flip-Flop RNN architectures for mapping oscillatory dynamics onto experimental signals, evaluating reconstruction in both time and frequency domains.
- Showed that recurrent and dynamical-systems-inspired architectures preserve the 13 to 30 Hz beta-band synchronization peak, where purely feedforward models lose this temporal structure entirely.

Computational Analysis of Memory Load in Sentence Comprehension Nov 2024 – May 2025

Cognitive Science Department, IIT Kanpur

- Built computational pipelines on multilingual Universal Dependencies data to extract structural features (dependency length, intervening elements, sentence complexity) and quantify their effect on memory load using linear mixed-effects models.
- Analysed how structural constraints shape information processing across languages, linking formal linguistic structure to processing difficulty. Resulted in a first-author preprint and two conference acceptances (see Publications).

Computational Behavioural Analysis: Systems Neuroscience Sept 2024 – Dec 2024

HM Lab, IISER Mohali

- Developed Python/OpenCV pipelines to extract behavioral variables from rodent video recordings in a systems neuroscience experiment, including motion tracking and pose-based feature extraction.

Scientific Skills

Dynamical Systems and Computational Modelling: Nonlinear dynamics; Coupled-oscillator systems; Bifurcation analysis; Biophysical neuron modelling; Recurrent network dynamics; Numerical simulation of nonlinear systems
Machine Learning & Deep learning: Convolutional and recurrent neural network architectures; Sequence and time-series modelling; Model evaluation and hyper-parameter tuning; Gradient boosting and classical ML pipelines
Programming and Tools: Python (NumPy, SciPy, Pandas, StreamLit, LangChain, PyTorch, TensorFlow, JAX); C++, MATLAB, Julia; Git, Unix/Linux, LaTeX.
Brain Signal Analysis: EEG and LFP processing; ICA-based artefact removal; Spectral and Phase-based methods; Epoching and event-locked analysis; Multivariate decoding.
Scientific Computing: Parallel processing; Reproducible computational workflows; Data visualization (Matplotlib, Seaborn); Linear mixed-effects models; Statistical inference & Hypothesis testing.

Education

IISER Mohali — **Jan 2022 – Ongoing**
Integrated BS–MS (Biology Major, Data Science Minor) | CPI: 8.6/10 — Expected Graduation: June 2026
Relevant Focus: Computational Neuroscience, Machine Learning, Dynamical Systems

Selected Computational & Theoretical Modelling Projects

Biophysical Neuron Simulation (Hodgkin–Huxley Model)
• Implemented and analysed conductance-based neuron models to study action potential generation, threshold dynamics, and ion channel contributions to spike timing.

Synchronization in Coupled Oscillators (Kuramoto Model)
• Simulated phase synchronization in coupled oscillator systems, studying transitions to coherence as a function of coupling strength and natural frequency distribution.

Spatio-Temporal Chaos in Coupled Map Lattices
• Studied pattern formation, traveling waves, and chaotic dynamics in nonlinear discrete systems with local coupling.

Motion Detection and Behavioural Analysis Pipeline
• Developed a computer vision-based pipeline using OpenCV to extract and analyze motion patterns from video data.

Relevant Coursework

Mathematics: Non-linear Dynamics, Linear Algebra, Probability & Statistics, Calculus, Differential Equations & Functions, Vector Calculus.
Machine Learning & Computer Science: Modelling Complex Systems, Machine Learning & AI, Data Structures & Algorithms, Biocomputing, Introduction to Data Science, Introduction to Computers.
Neuroscience & Biology: Neuroscience, Computational Structural Biology, Bioinformatics, Animal Physiology, Biostatistics & Hypothesis Testing, Genetics & Genomics.